CUSTOMER SEGMENTATION:
A KEY TO UNLOCKING BUSINESS GROWTH AND SUCCESS

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Machine Learning II

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1 Abstract

This project applies **unsupervised machine learning** to **segment customers** based on demographic, behavioral, and transactional data. Through **clustering** techniques and **association rule mining**, distinct customer profiles were identified and enriched with purchasing patterns. These insights support **targeted marketing strategies** and **data-driven decision-making**. A **spatial cluster** was also manually defined to capture **geographic behavior**. The approach demonstrates the value of **segmentation** in improving customer understanding and promotional effectiveness.

Keywords: customer segmentation, clustering, unsupervised learning, association rules, marketing strategy, consumer behavior.

2 Introduction

Understanding **customer behavior** is pivotal for developing effective **marketing strategies** and maintaining a **competitive advantage** in today's dynamic market landscape. Businesses that can **segment** their customers into **meaningful groups** gain the ability to tailor their marketing efforts, improve customer engagement, and ultimately **drive growth**. This project focuses on leveraging **data-driven techniques** to perform **customer segmentation** by analyzing demographic, behavioral, and transactional data.

2.1 Problem Statement

As markets grow increasingly saturated and **consumer preferences grow more diverse**, businesses face challenges in effectively targeting and retaining customers. Traditional one-size-fits-all marketing approaches often fail to address the unique needs of different customer groups, resulting in **lower engagement** and **missed opportunities**. The core problem addressed in this project is how to segment a heterogeneous customer base into actionable clusters that reflect shared characteristics and behaviors. These segments will enable more personalized and effective marketing campaigns that resonate with customers' distinct needs.

2.2 Role of Unsupervised Learning

Unsupervised learning techniques provide robust tools for discovering **hidden patterns** in data without relying on pre-labeled examples. These methods enable the identification of intrinsic structures and relationships within customer data, facilitating the extraction of meaningful segments. Techniques such as **clustering** reveal underlying similarities and differences among customers, empowering data-driven decision-making and insightful customer profiling.

2.3 Explain Datasets

Two main datasets were used, each supporting different parts of the analysis.

- Customer Information Dataset

This dataset provides detailed **demographic** and **spending data** for each customer, including personal details, location, lifetime spend across product categories, store visits, and shopping behavior. It enables the creation of **customer segments** through clustering based on their profiles and purchase habits.

- Customer Basket Dataset

This dataset contains **transaction-level data** with lists of products purchased in each basket, linked to customers via customer IDs. It is used to identify **association rules** within each segment, revealing which products are frequently bought together.

By combining these datasets, we segment customers into meaningful groups and then use their transaction patterns to design **targeted promotions and marketing strategies** tailored to each segment.

3 Methodology

This section details the methodological framework applied in the project, from initial data exploration to advanced preprocessing steps. It highlights the **technical procedures** used to prepare the data for analysis, including cleaning, transformation, and feature selection, ensuring that the data is suitable for clustering algorithms.

3.1 Data Exploration and Preprocessing

3.1.1 Data Exploration

The *customer_info* dataset consists of 34060 observations and 26 features. It contains **demographic, behavioral, transactional, and geographic attributes**. While some features show a wide range of unique values (indicating high variability), others appear limited in scope. Certain values related to purchasing behavior and geographic location exhibit **high uniqueness**, suggesting they **offer strong individual-level differentiation**. Conversely, household composition and complaint-related fields present lower variability.

3.1.2 Data Cleaning

In the data cleaning step, several actions were carried out to **improve the quality and usability** of the dataset.

3.1.2.1 Unnamed Column

First, an **unnamed column**, typically generated when exporting or importing data from CSV files, was **removed**. This column did not contain any relevant information and was therefore considered **redundant**.

3.1.2.2 Set customer_id as index

Next, the ***customer_id*** column was set as the **index** of the dataset. This was done to uniquely identify each record based on the customer and to facilitate more efficient data manipulation.

3.1.2.3 Standard Missing Values

Lastly, all occurrences of the **string "NaN"** were replaced with pandas' standard missing value indicator, ***pd.NA***. This replacement ensures **uniform handling of missing data** throughout the analysis and supports accurate application of pandas' built-in functions for data cleaning and imputation.

3.1.3 Manual Cluster

As part of the exploratory phase, we analyzed the **latitude and longitude coordinates** of all customers to assess whether **any spatially coherent clusters** could be identified manually. The objective was to explore whether **geographic concentration alone** could reveal meaningful behavioral or commercial groupings that might not be captured by conventional clustering algorithms.

3.1.3.1 Geospatial Visualization

Customer coordinates were plotted using **multiple visualization techniques**, including static map overlays with OpenStreetMap basemaps (***Figure 1***), interactive cluster maps, and density-enhanced heatmaps. These views highlighted the presence of **spatially concentrated customer groups**.

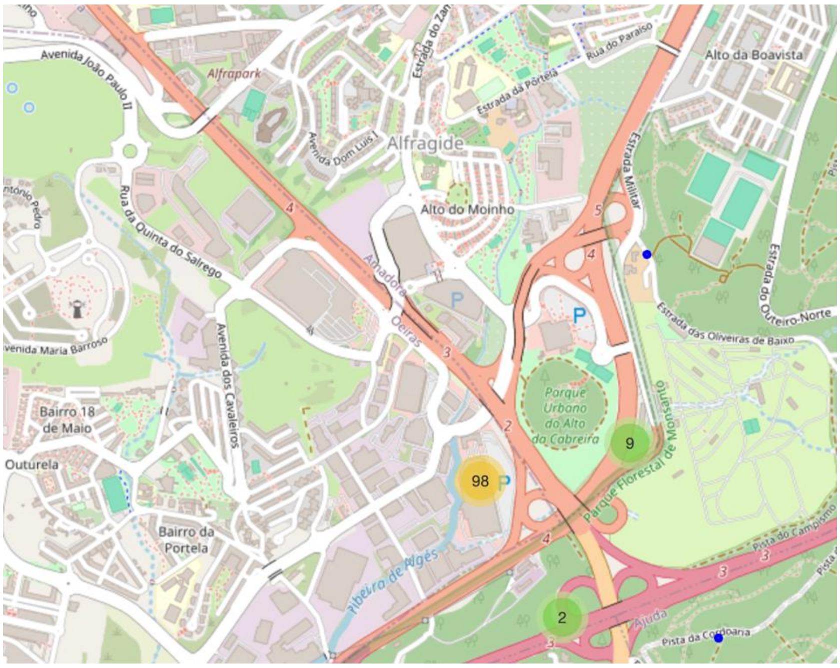


Figure 1- OSM

<u>Note:</u> The **yellow-colored circle** represents the **Makro Alfragide cluster**, where 98 customers are shown. Later, our manual delimitation identified **97 individuals**, and this minor discrepancy is expected due to the **spatial approximation inherent in geographic visualizations**.

3.1.3.2 Density-Based Inspection

Kernel Density Estimation (KDE) was used to accentuate regional density differences. This enabled visual confirmation of meaningful geographic clusters beyond mere visual noise.

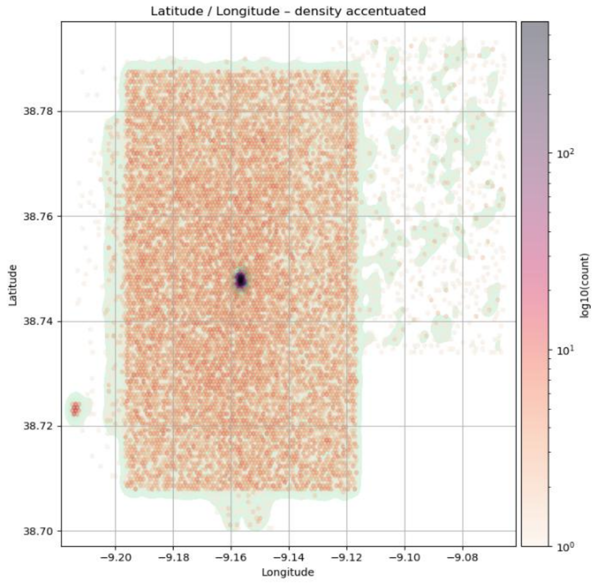


Figure 2- KDE

3.1.3.3 Cluster Delimitation

Based on the observed spatial patterns, **manual geographic bounds** were defined to isolate the dense cluster of customers in the **Makro Alfragide region**. Specifically, a **bounding box** was created using the following coordinate thresholds: **longitude between -9.2150 and -9.2125**, and **latitude between 38.7220 and 38.7245**. Customers falling within this range were flagged as members of a distinct **makro-cluster**, representing a geographically informed segmentation. The coordinate thresholds were determined through an interactive map visual inspection of the spatial distribution (***Figure 3***).

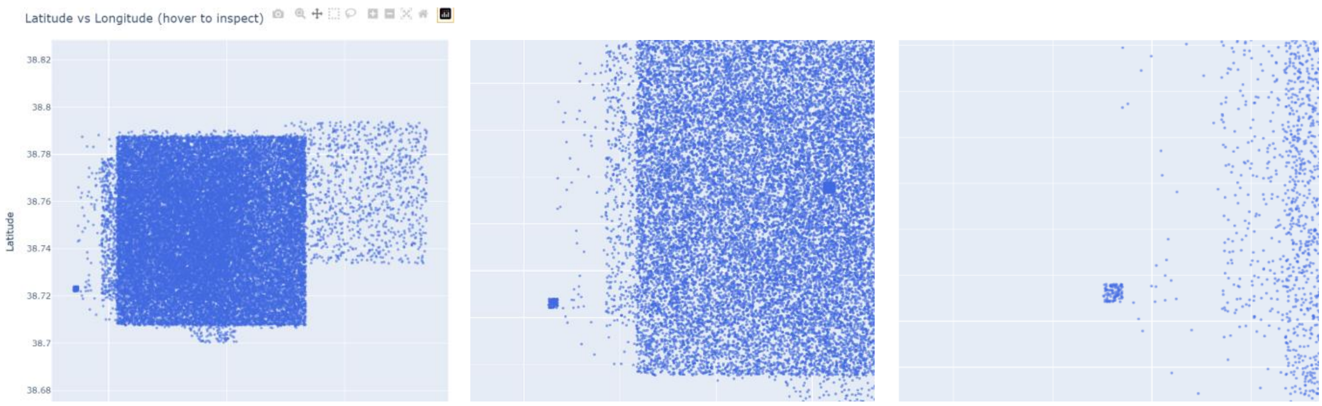


Figure 3 – Bounding Box

3.1.3.4 Segregation of the Cluster

The identified group, consistent of 97 customers, was isolated into a separate dataset *makro_cluster* for dedicated analysis. The remaining 33963 records constituted the primary dataset for general clustering tasks.

This exploratory spatial segmentation offered a **real-world, interpretable baseline** for understanding regional customer behaviors and evaluating whether location-driven patterns aligned with behavior-based clustering results.

3.1.4 Inconsistencies

During data preprocessing, it is crucial to identify and address **inconsistencies** that may compromise the integrity of the analysis. Inconsistencies refer to values that fall outside their expected ranges, violate logical constraints, or do not align with real-world conditions.

3.1.4.1 Percentage of Products Bought on Promotion

The feature representing the percentage of products brought on promotion should logically be bounded between **0 and 1**, where 0 indicates no purchases were made on promotion, and 1 indicates all purchases were promotional. Values falling outside this range may result from data entry errors or flawed calculations. To maintain data validity, all values in this column were **constrained** to fall within the [0,1] interval.

3.1.4.2 Year of First Transaction

Another inconsistency involved the year of the customer's first recorded transaction. Since this value should reflect a **past or current event**, any instances where the transaction year was later than the current year were flagged as invalid. Such entries suggest either data entry mistakes or system errors and were identified for correction or removal to ensure temporal consistency within the dataset.

3.1.5 Visual Exploratory Analysis

Univariate plots were generated for all numerical features to assess data distribution and **identify missing values** (***Annex1***). Each plot included a density curve and a count of missing entries, allowing for quick evaluation of both completeness and statistical shape.

Several features showed noticeable levels of missingness. The visualizations also revealed varying distribution types, which informed appropriate imputation strategies in later preprocessing steps.

3.1.6 Removing Constant, Semi-Constant Features, and Duplicates

Diagnostic checks were performed to identify **constant** and **semi-constant columns**, as well as **duplicate observations.** None were detected. Nonetheless, this step remains a crucial component of the preprocessing pipeline in clustering tasks, as such observations can introduce noise, diminish model efficiency, and distort similarity measures.

3.1.7 Encoding Categorical Variables

The dataset contained a **single categorical variable: *gender***. To convert this variable into a numerical format suitable for analysis, a binary encoding scheme was applied. Specifically, the values were **mapped** as follows: *Male* was encoded as 0, and *Female* as 1.

3.1.8 Feature Engineering

3.1.8.1 Age Column

An **age variable** was derived from the **customers' birthdate** information to provide a more interpretable and continuous measure of age. This transformation allows for a more meaningful representation of customer demographics and removes the need for storing raw birthdate data, which can be less practical for analysis. Following this transformation, the original birthdate column was removed to eliminate redundancy.

3.1.8.2 Cyclic Variables: Typical Hour

The typical hour variable, representing the time of day associated with customer activity, was transformed using **sine and cosine functions** to capture its **inherent cyclic nature**. This encoding preserves the periodicity of hours within a 24-hour cycle. After this transformation, the original column was dropped, as its information was fully represented by the new cyclic features.

3.1.8.3 Add Loyalty Flag

A **binary loyalty flag** was introduced to indicate whether a **customer participates in the loyalty program**, with customers holding a loyalty card assigned a value of 1. This feature replaced the original loyalty card number column, which was subsequently removed to simplify the dataset and focus on relevant behavioral indicators.

3.1.8.4 Add Years of Education

Years of education were derived as a **quantitative feature** representing customer educational attainment. To avoid redundancy and remove less informative dates, the customer's name column was dropped following this transformation.

The years were assigned as follows:

- Phd: 22 years
- Master: 17 years
- Bachelor: 15 years
- Unknown/High-School: 12 Years

3.1.8.5 Add Years since First Transaction

A feature quantifying the **number of years since a customer's first transaction** was created to better capture customer tenure and engagement. The original year of the first transaction column was removed after this derivation to streamline the dataset.

3.1.8.6 Lifetime Spend Technology

Two highly correlated variables - lifetime expenditures on electronics and **video games** (as further examined in Section 3.2) - were combined into a single composite feature representing the **lifetime spending on technology-related products**. Both original columns were dropped to reduce dimensionality while maintaining conceptual clarity in spending behavior representation.

3.1.9 Outlier Detection and Removal

To ensure a robust clustering process and minimize the influence of extreme or anomalous data points, a structured **outlier detection** strategy was employed. As a first step, the dataset was divided into two groups of features based on their nature:

- **Bounded variables**: features with a natural or business-defined upper limit (e.g., number of children or complaints).
- **Continuous variables**: features without explicit bounds, requiring statistical detection of anomalies.

3.1.9.1 Bounded Variables - Threshold Capping

For bounded variables, a threshold capping procedure was applied to **restrict values to plausible limits**. This step aimed to prevent unrealistic outliers from **skewing** the analysis. The thresholds were defined using domain knowledge and applied uniformly across the dataset:

- *kids_home* was capped at **3** (2,374 entries affected),
- *teen_home at* **2** (962 affected),
- *number_complaints* at **2** (680 affected),
- *distinct_stores_visited* at **6** (570 affected).

This preprocessing step ensured that extreme values in constrained features did not distort similarity measures in clustering.

3.1.9.2 Continuous Variables - Univariate and Multivariate Outlier Detection

For continuous features, a multi-method approach was used to identify outliers from both univariate and multivariate perspectives:

- **Interquartile Range (IQR) Method**: A univariate statistical method was used to flag observations falling beyond 1.75 times the IQR from the first or third quartile. This method is effective in detecting isolated feature-level anomalies.
- **DBSCAN (Density-Based Spatial Clustering)**: This multivariate technique identifies observations in low-density regions based on a fixed neighborhood radius and minimum point count. Points outside dense clusters are considered potential outliers.
- **Self-Organizing Maps (SOM)**: SOMs were trained on scaled, complete data to calculate the quantization error for each record. Observations with error above a defined high-percentile threshold were flagged as multivariate anomalies.

3.1.9.3 Consensus-Based Filtering

To adopt a conservative and reliable filtering approach, only records identified as outliers **by all three methods** - IQR, DBSCAN, and SOM - were removed. This consensus-based logic **minimizes false positives** and ensures only consistently anomalous records are excluded.

3.1.9.4 Control on Data Loss

To avoid excessive reduction in sample size, a safeguard was introduced: if the number of records flagged for removal exceeded **5%** of the dataset, the process was aborted. Conversely, if too few records were flagged, a warning was issued to reassess thresholds.

3.1.9.5 Preservation of Removed Records

Rather than discarding the filtered records, all identified outliers were stored in a **separate dataset** (*outlier_dataset*). These records will be later assigned to the closest cluster centroid using post-clustering distance calculations, allowing for their inclusion in profiling without influencing the clustering model itself.

3.1.10 Feature Scaling and Handling Missing Values

In preparation for clustering, a structured scaling and imputation workflow was applied to address differences in feature magnitude and the presence of missing values.

3.1.10.1 Multiple Scaling Strategies

To evaluate the impact of scaling on clustering performance, three versions of the dataset were created using different scalers:

- **StandardScaler** (mean = 0, variance = 1),
- **RobustScaler** (resistant to outliers via median and IQR),
- **MinMaxScaler** (scales to [0, 1] range).

Each scaled dataset was evaluated independently to determine which preprocessing pipeline yielded better clustering performance. **Binary columns were not scaled**, as their discrete nature and fixed range (0 or 1) do not benefit from standard scaling techniques and could distort the interpretation of distance-based algorithms.

3.1.10.2 Feature Categorization

To streamline the pipeline, features were categorized into three types:

- **Tailed features**: continuous variables prone to outliers (e.g., monetary or duration-based variables)
- **Bounded features**: constrained numeric features with logical upper limits (e.g., number of kids, number of stores)
- **Excluded features**: binary variables (e.g., gender, loyalty program flag) and geographic coordinates (latitude and longitude), which were excluded from clustering. Binary features tend to distort distance-based metrics, and geographic variables were considered semantically irrelevant to behavioral segmentation

3.1.10.3 Initial Evaluation

A preliminary model evaluation was conducted across the three scaled datasets using only the **clustering features** (i.e., bounded + tailed). For each dataset, we tested two imputation strategies:

- **Median imputation**
- **K-Nearest Neighbors (KNN)** with *n_neighbors=5*

These combinations were tested using:

- **K-Means** with a grid search for optimal cluster number,
- **Hierarchical Clustering** using Ward linkage,
- **Self-Organizing Maps (SOMs)** with fixed grid dimensions.

Additionally, a **hybrid imputation strategy** was tested using the **Robust-scaled dataset**, where:

- Tailed features were imputed using **median**, and

- Bounded features were imputed using **KNN (k=5)**.
- **DBSCAN and Mean Shift Not Applied**

While DBSCAN and Mean Shift were initially considered, these methods were not applied due to their known sensitivity to varying feature densities. The high dimensionality and heterogeneous scale of our dataset make these algorithms less suitable, as they tend to struggle with non-uniform cluster structures.

3.1.10.4 Selection of Final Preprocessing Pipeline

Across all tested clustering methods, the **RobustScaler with hybrid imputation** consistently outperformed other combinations. Based on these findings, this configuration was selected as the final preprocessing pipeline for clustering.

3.1.10.5 Outlier dataset preparation

The **outlier dataset**, excluded from model training, was preprocessed separately to enable its integration into the final segmentation. As with the main dataset, the **RobustScaler** was applied (**except on the binary columns**) to ensure consistent feature scaling. Regarding missing values, only the cyclic variables derived from *typical_hour* contained gaps. Since these were continuous and bounded between -1 and 1, a KNN imputation (k=5) strategy was applied uniformly. The parameter k = 5 was selected as it produced satisfactory results in initial tests and corresponds to the default setting in most standard implementations, offering a reasonable balance between local accuracy and generalization. A broader search across different *k* values was considered unnecessary, as the computational cost would likely outweigh the marginal improvements in imputation quality, particularly given the limited influence of the missing values on overall clustering performance.
This light preprocessing ensured the outliers were properly aligned with the clustering features and could be later assigned to the most appropriate segment via nearest-centroid methods.

3.2 Feature Selection

In the segmentation task, selecting the most informative numerical features is essential to enhance model performance and interpretability. To achieve this, two key criteria guided our approach.

3.2.1 Understanding Variance

Assessing the variance within features helps identify those that **provide meaningful differentiation across observations**, ensuring the selected features contribute valuable information to the analysis.

3.2.2 Correlation Matrix

To recognize the most relevant features, a **correlation matrix** was employed. This matrix enabled a comprehensive examination of the strength and direction of relationships between features.

Thanks to insights from the correlation matrix, we identified a **strong positive correlation between** lifetime expenditures on video games and electronics (previously detailed in Section 3.1.8.6 Lifetime Spend Technology). To capture this relationship effectively while reducing dimensionality, these two variables were combined into a single composite feature representing lifetime spending on technology-related products.

After completing all preprocessing and feature engineering steps, the finalized dataset containing only the selected and transformed features was exported as '*info_clustering.csv*' for use in the subsequent clustering analysis.

4 Clustering and Customer Segmentation

Clustering is the core analytical technique used to segment customers based on similarities in their behavior and attributes. This section explores various clustering methods, the process for selecting the optimal model, and the insights derived from the resulting customer segments.

4.1 Model Selection and Optimal Number of Clusters

4.1.1 K-Means

To begin the clustering process with K-Means, the preprocessed dataset ***(info_clustering.csv*)** was imported. In accordance with the segmentation strategy defined earlier, all **binary variables** (e.g., gender, loyalty flag) and **geolocation features** (latitude and longitude) were **excluded from the clustering procedure**.

4.1.1.1 Determining the Optimal Number of Clusters

To determine the optimal number of clusters for the K-Means algorithm, both the **Elbow Method** and **Silhouette Score** were computed using a custom function that evaluated clustering performance for values of *k* ranging from 2 to 10. For each *k*, the function recorded the model's **inertia** (within-cluster sum of squares) and the **average silhouette score**, providing a dual perspective on model cohesion and separation.

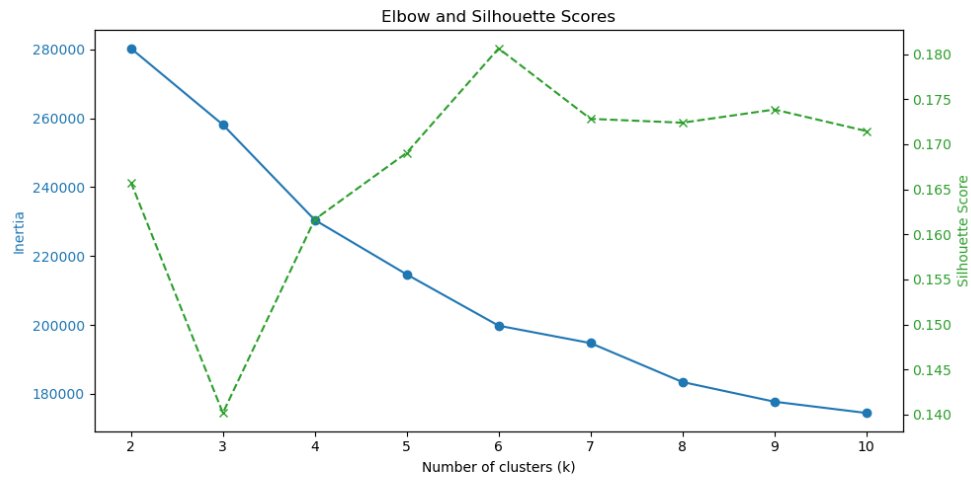


Figure 4- Elbow and Silhouette Scores

The results (***Figure 4***) revealed that **k = 6** simultaneously exhibited a **clear elbow shape** in the inertia curve and achieved the **highest silhouette score** of **0.1806**, thus supporting the selection of six clusters as the optimal solution.4

Before going to the next step, cluster sizes were examined to assess the **distribution of customers across the six segments**. The analysis confirmed that the clusters were **generally well balanced**, apart from **Cluster 2**, which had a noticeably bigger size. However, this was deemed acceptable given its **distinct behavioral characteristics**, which justified its presence as a standalone segment within the overall solution.

4.1.1.2 Centroid Extraction

Following the selection of k = 6, the final K-Means model was trained on the dataset excluding binary and geolocation features. The resulting **cluster centroids** were extracted to represent the average feature profile of each cluster. A DataFrame was then created with the corresponding **feature names** and assigned cluster labels *(kmeans_cluster)* for clarity. This centroid matrix served as the foundation for subsequent **cluster interpretation**.

4.1.1.3 Visualizing Cluster Structure with UMAP

To gain insight into the spatial distribution and separability of the six K-Means clusters, **Uniform Manifold Approximation and Projection (UMAP)** was applied for dimensionality reduction. The clustering features were used as input, and the UMAP model was configured with n_neighbors=15 and min_dist=0.1 to balance local and global structure. The resulting 2D embedding was visualized using a custom plotting function, with points colored by their assigned cluster label. This visualization provided an intuitive overview of cluster cohesion and overlap in reduced-dimensional space.

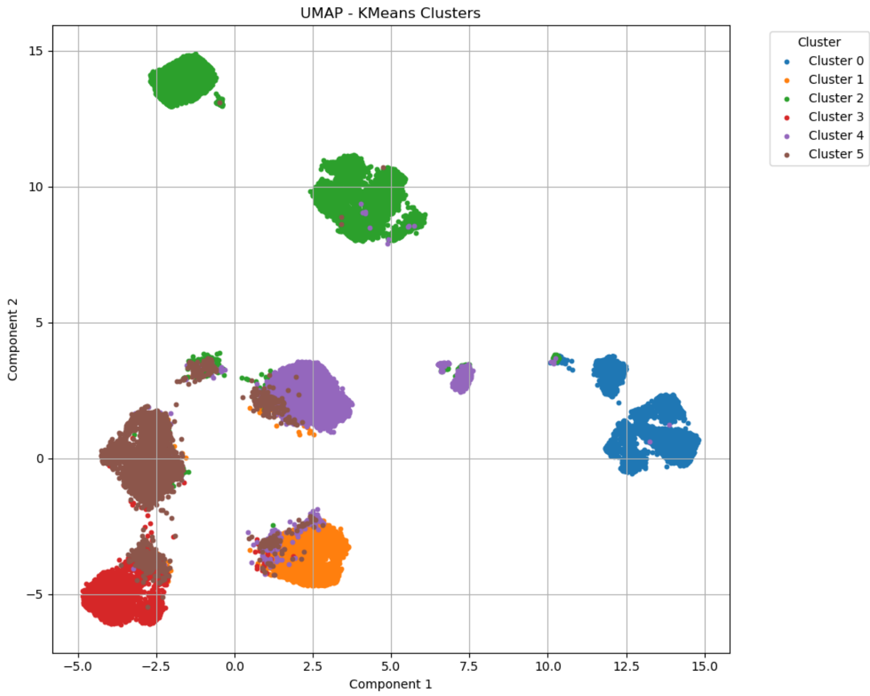


Figure 5- UMAP, KMeans Clusters5

4.1.1.4 Visualizing Cluster Structure with PCA

To complement the UMAP visualization and provide a **linear projection** of the clustering structure, **Principal Component Analysis (PCA)** was applied to the same feature set used for clustering. The data was reduced to two principal components, capturing the most significant directions of variance. A custom visualization function was used to plot the resulting 2D embedding, with points colored according to their assigned **K-Means cluster**. This plot offered an interpretable overview of how well the clusters separate along the most informative linear dimensions.

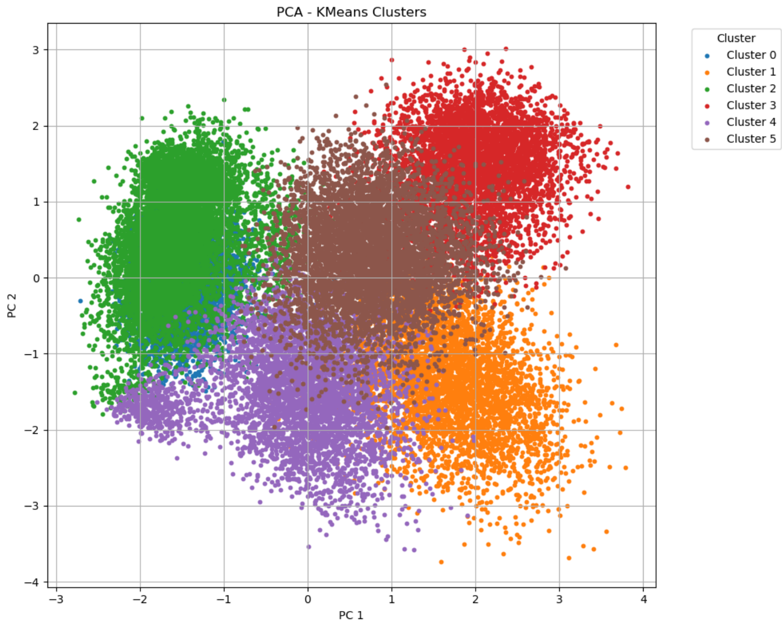


Figure 6- PCA, KMeans Clusters6

4.1.1.5 Silhouette Analysis

To assess the internal consistency and separation of the K-Means clustering solution, a **silhouette analysis** was performed. Using the same feature set applied during model training, the **silhouette coefficient** was computed for each data point, providing a measure of how similar each observation is to its own cluster relative to other clusters. A custom visualization function was used to plot the silhouette scores for all clusters.

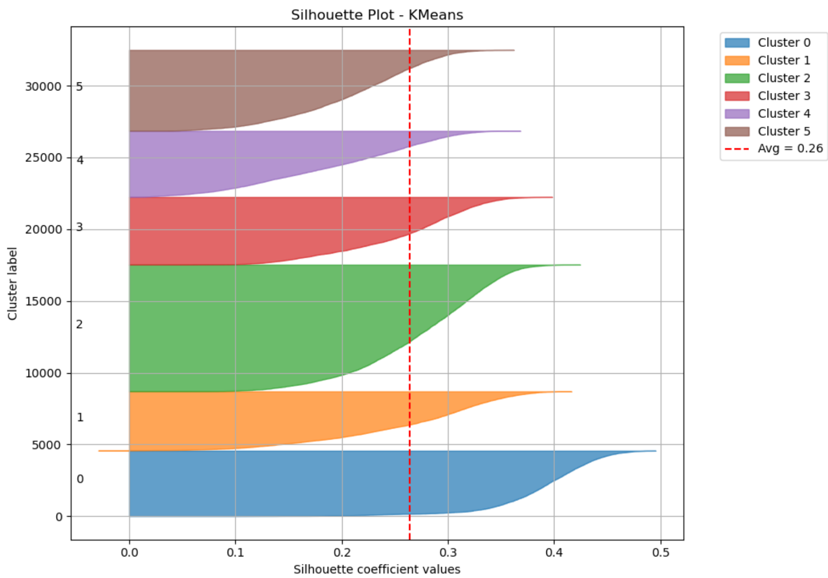


Figure 7 – Silhouette Plot, KMeans7

The average silhouette score was **0.26**, indicating a **moderate level of cohesion and separation** across the six clusters. As visualized in ***Figure 7***, **Cluster 0** and **Cluster 2** exhibited stronger internal consistency, with most data points showing relatively high silhouette values. While some clusters displayed greater internal variability, the overall clustering structure was considered

sufficiently robust for exploratory purposes. Based on these results, and to further validate the segmentation findings, we proceeded to test an **alternative clustering approach**

4.1.2 Hierarchical Clustering (with KNN centroids)

4.1.2.1 Hierarchical Clustering Initialization

To explore alternative segmentation structures, we applied **Hierarchical Clustering** using a **K-Means initialization with *k* = *20***. This approach allowed us to **generate initial cluster labels** via K-Means offering a computationally efficient starting point before proceeding with hierarchical techniques. The preliminary K-Means model at *k* = *20* achieved a **silhouette score of 0.119**, providing an initial benchmark for the cluster quality. This step ensured a structured basis for further refinement and linkage-based clustering.

The cluster labels obtained from the K-Means initialization were stored in the dataset under the column '*kmeans20_cluster'*, serving as the initial segmentation for the subsequent hierarchical clustering process.

4.1.2.2 Storing K-Means Centroids for Hierarchical Initialization

The resulting **centroids** were extracted and stored in a DataFrame, with each row representing the average feature profile of one of the 20 clusters. These centroids were labeled using the kmeans20_cluster identifier to facilitate interpretation and traceability. This intermediate segmentation provided a structured and efficient starting point for the subsequent **macro-cluster formation through hierarchical linkage analysis**.

4.1.2.3 Hierarchical Clustering on K-Means Centroids

To derive higher-level groupings from the 20 K-Means clusters, **agglomerative hierarchical clustering** was performed on the corresponding centroids using **ward's linkage method**. The resulting dendrogram (***Figure 8***) was used to determine an appropriate number of macro-clusters. While both *k* = *5* and *k* = *6* appeared viable based on visual inspection, we selected **k = 6** to maintain consistency with the K-Means segmentation and to allow for a more nuanced interpretation of the underlying structure. Additionally, experimental testing revealed that *k* = *5* produced **inferior cluster cohesion**, supporting the choice of six macro-clusters for final interpretation.

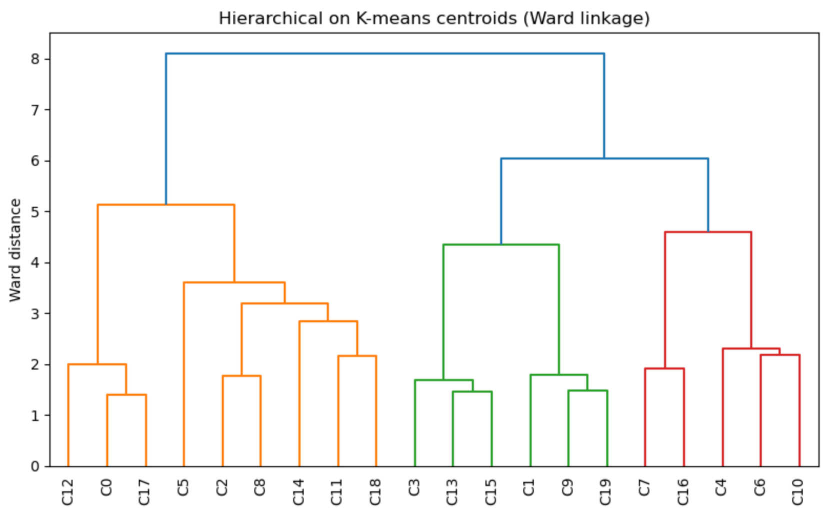


Figure 8 – Dendogram of Hierarchical Clustering8

4.1.2.4 Assigning Macro-Cluster Labels to Customers

After selecting $k = 6$ as the number of macro-clusters, the corresponding labels were extracted from the dendrogram using the *fcluster* function, resulting in six distinct groupings of the original 20 K-Means centroids. These labels were then mapped back to all customers based on their initial kmeans20_cluster assignment. The new macro-cluster identifiers were stored in the column macro_cluster, enabling downstream analysis at a more aggregated segmentation level. To assess the distribution of customers across the newly assigned macro-clusters, **value counts were computed again**, providing a clear view of cluster sizes and ensuring that the segmentation did not produce highly imbalanced groupings.

4.1.2.5 UMAP Visualization of Hierarchical Segmentation

To visually assess the structure and separability of the macro-clusters derived from hierarchical clustering, **UMAP** was applied to the same feature set previously used for K-Means clustering. The model was configured with n_neighbors = 15 and min_dist = 0.05 to preserve both local and global relationships within the data. The resulting 2D embedding (***Figure 9***) revealed **clearly separated and compact cluster formations**, indicating strong internal cohesion and distinct segment boundaries. This visualization provided qualitative support for the validity of the six-group hierarchical solution.

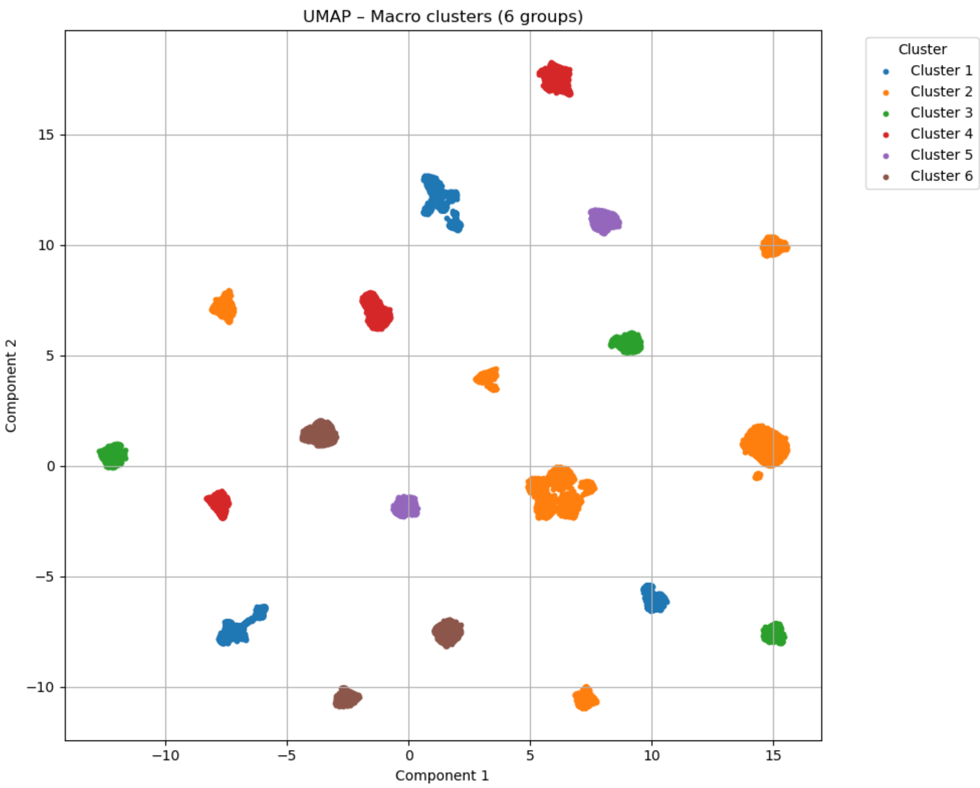


Figure 9- UMAP, Macro Cluster9

4.1.2.6 Silhouette Analysis of Macro Clusters

To evaluate the cohesion and separation of the six macro-clusters derived from hierarchical clustering, a **silhouette analysis** was conducted using the same set of clustering features. The resulting silhouette plot (***Figure 10)*** provided a visual assessment of the internal consistency of each group.

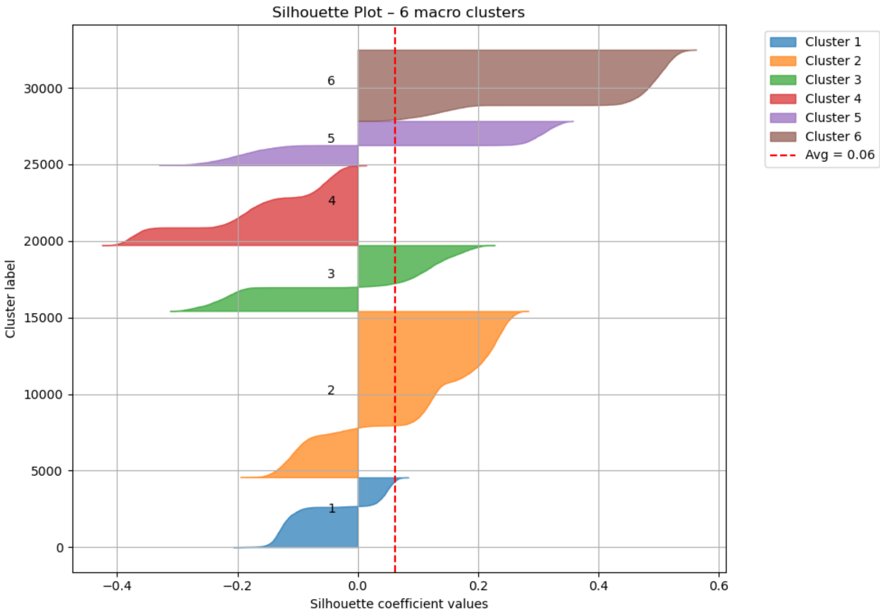


Figure 10 – Silhouette Plot – 6 Macro Clusters10

The average silhouette score was 0.06, which, although lower than that observed in the K-Means solution, still suggests a **basic underlying structure** in the data. As illustrated in Figure 10, Clusters 5 and 6 demonstrate strong internal cohesion, with most of their points showing positive silhouette values, indicative of compact and well-defined groupings. In contrast, Clusters 1, 2, and 4 exhibit greater variability and overlap, with many data points registering low or even negative silhouette scores. This pattern points to weaker separation and potential ambiguity in cluster boundaries. Overall, although the macro-clusters offer a meaningful high-level segmentation of customer behavior, the silhouette analysis suggests that some segments may be **less distinctly defined** and should therefore be interpreted with caution in subsequent profiling.

4.1.2.7 Post-Clustering Assignment of Outliers to K-Means Segments

Since the K-Means solution demonstrated superior performance compared to the hierarchical segmentation, we proceeded to assign the previously excluded **outliers** to the clusters defined in the **K-Means model**. To enable this, the outlier dataset was imported into the clustering notebook and prepared for post hoc cluster assignment using the centroids from the K-Means segmentation.

4.1.2.8 Assigning Outliers via Nearest Centroid Method

To integrate the outliers into the existing segmentation framework, **Euclidean distances** were calculated between each outlier and the centroids of the six K-Means clusters. The *pairwise_distances_argmin_min* function was employed to assign each outlier to its **nearest**

centroid, based on minimum distance. The corresponding cluster labels were stored in a new column, *kmeans_cluster*, within the outlier dataset. This procedure ensured that all records both inliers and outliers were consistently mapped to the original K-Means segmentation for unified analysis.

4.1.2.9 Merging Outliers with the Final Clustered Dataset

Finally, the **original clustered dataset** and the **assigned outlier records** were concatenated to form a unified dataset *(dataset_cluster)*, enabling comprehensive analysis across all customer segments, including those initially excluded due to outlier status.

To preserve the complete segmentation output, including both core and outlier records, the unified dataset was exported as *dataset_cluster_final.csv*.1

4.1.3 Insights from Self-Organizing Maps

The application of Self-Organizing Maps (SOMs) in this study aimed not at direct clustering, but rather at **uncovering meaningful patterns and relationships within the dataset**. A high-resolution **SOM grid of 30x30 neurons** was employed. This intentionally large configuration was chosen to enhance the granularity of visualization, thus facilitating a more nuanced interpretation of the data's underlying structure.

4.1.3.1 Training Evaluation via Hit Map

To assess the effectiveness of the training process, a **hit map** was generated. In this map, each neuron is assigned a value corresponding to the number of input samples for which it served as the **Best Matching Unit (BMU).** Darker areas represent neurons that attracted more samples, while lighter areas indicate less frequently matched neurons.

The resulting hit map indicates a well-balanced SOM: there are no excessively dark or light regions, suggesting that the training process was well-optimized. Each neuron has successfully adapted to represent a distinct subset of the data, with minimal evidence of overfitting or underrepresentation.

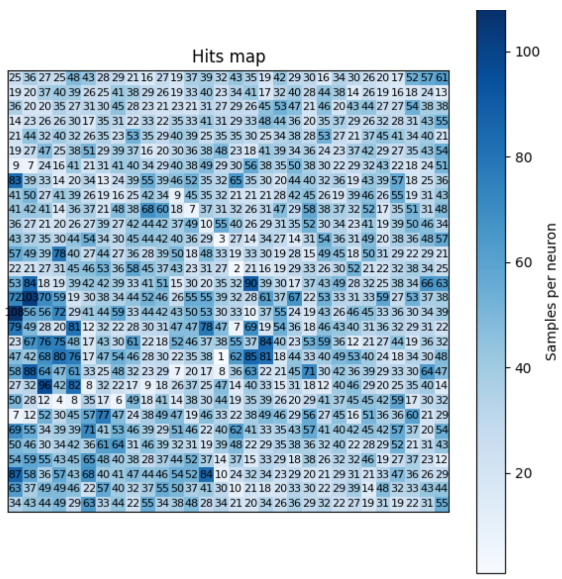


Figure 11 – Hit Map

4.1.3.2 Feature Map Analysis

Feature maps were also plotted for each variable. In these maps, the color intensity of each neuron reflects the weight value associated with that variable. This visualization helps identify spatial patterns and correlations among features across different segments of the data.

The most notable insights from the feature maps are discussed below.

- Household Composition

Overlapping **red zones in the upper region** of the SOM grid correspond to **larger families**, while the **lower region** indicates households with **fewer or no children**. This spatial separation suggests a clear segmentation based on family size.

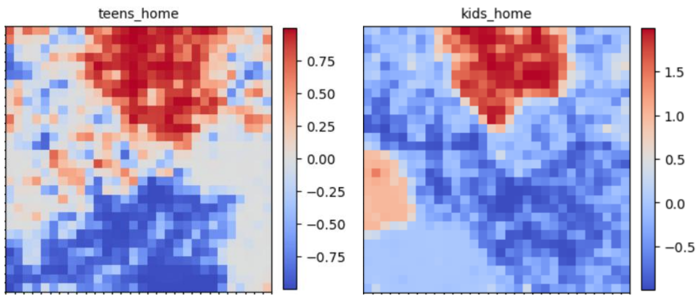


Figure 12 – Household Composition

- Groceries Spending Patterns

In the lifetime spending maps for **meat and fish**, the **bottom-left** region of the SOM grid shows a distinct group with **low expenditure on these items**. Conversely, the same region in the **vegetable spending map** reveals **high expenditures**, indicating a possible **cluster of vegetarian or vegan** consumers.

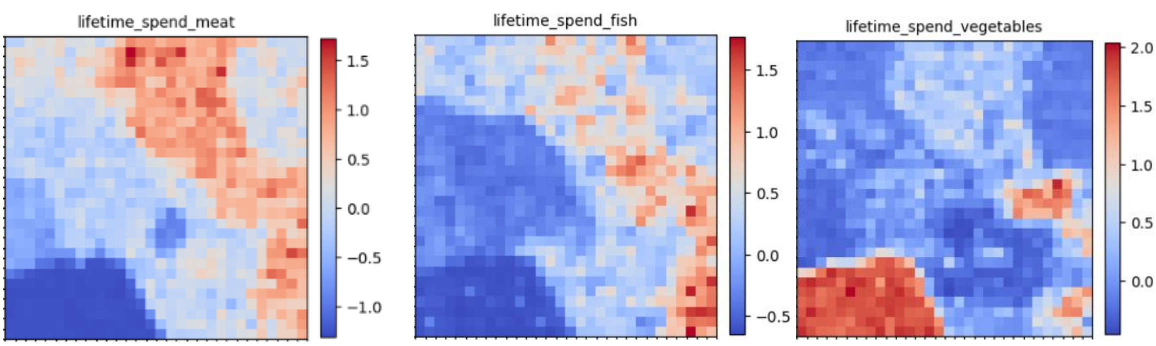


Figure ~~12~~13 – Groceries Spending Patterns

- Alcohol and Meat Consumption

A strong spatial overlap is observed **between the red zones** of the *lifetime_spend_alcohol_drinks* and *lifetime_spend_meat* maps. This suggests that customers who **frequently purchase alcoholic beverages** also tend to **spend more on meat**, revealing a potentially correlated consumer segment.

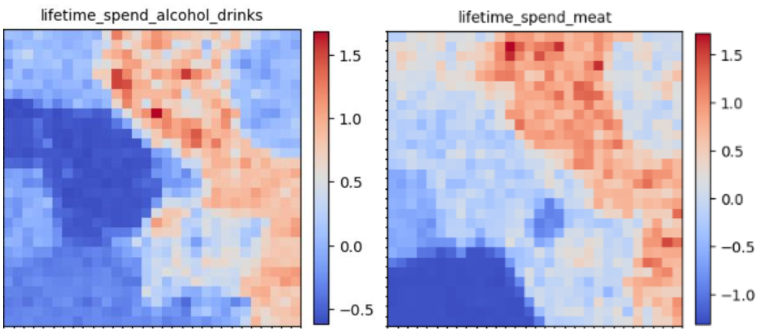


Figure ~~13~~14 – Alcohol and Meat Consumption

- Promotion Usage and Education Level

On the **left side** of the grid, a **higher proportion of products purchased through promotions** is observed. This area coincides with the **lowest values in the *years_education* map**. This inverse relationship implies that customers with fewer years of education are more likely to seek out promotional deals. This pattern may reflect budget-conscious behavior, often associated with lower-income segments. Additionally, this group may also include students, who typically have limited purchasing power and actively seek out discounts as a means of managing their expenses.

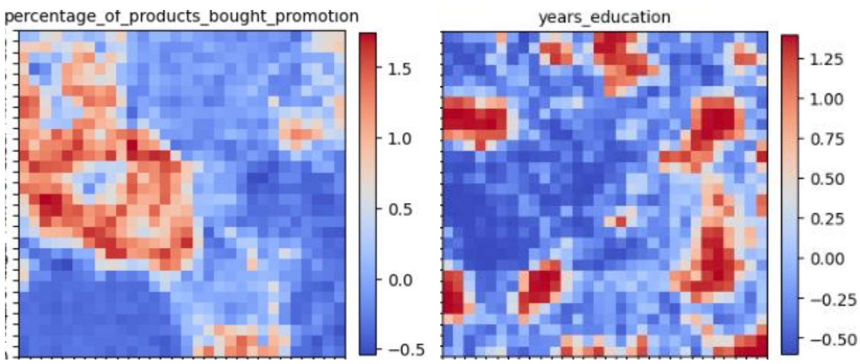


Figure 15 – Promotion Usage and Education Level

- Young Parents with High Complaint Rates

By examining the feature maps for *number_complaints*, *age*, and *number_children_at_home*, a noteworthy cluster emerges. On the **left side** of the SOM grid, a group is identified that reports a **high number of complaints**, consists of **younger individuals**, and has **children at home**. This segment likely represents young parents who are more vocal about their concerns and experiences with the service or products offered.

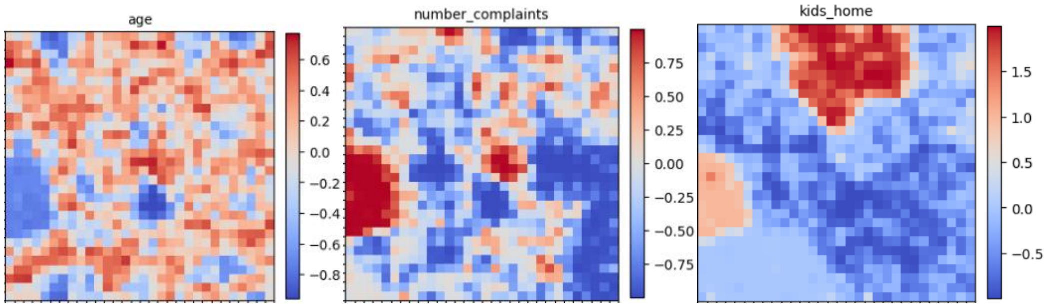


Figure ~~15~~16 – Young Parents and High Number of Complaints

4.1.4 Other Clustering Techniques

Alternative clustering methods such as **Mean Shift, DBSCAN, and K-Medoids** were not employed in this study. K-Medoids was excluded due to its **limited scalability and underdeveloped optimization** for large or complex dataset. DBSCAN and Mean Shift were also not suitable given the **high dimensionality of the data**, which negatively impacts their performance and effectiveness.

4.2 Clustering Results

For visualization purposes only, **UMAP and PCA** were utilized to represent high-dimensional data in two-dimensional space. UMAP was selected over t-SNE due to its **faster runtime**, **scalability, and ability to preserve both local and global data structure**, enabling more informative visualizations. PCA was employed to provide a linear, interpretable projection of variance. These visualizations supported cluster interpretation.

4.3 Cluster Profiling and Interpretations

4.3.1 Cluster 0 - Vegetarian

This group consists primarily of younger individuals who **spend very little on categories such as meat, fish, and pet food (Annex 10, Annex 12, Annex 13)**, and spend a lot on vegetables (**Annex 7**) potentially reflecting **vegetarian or plant-based preferences**. They typically shop during the afternoon (**Annex 20**), focus on purchasing a small number of essential items (**Annex 6**), and **rarely respond to promotional offers (Annex 15)**. Their minimalist and selective shopping behavior may indicate a lifestyle guided by ethical or health-conscious choices.

4.3.2 Cluster 1 - Wellness Urbanites

This cluster includes highly educated (**Annex 17**) and balanced consumers **who maintain consistent spending across categories such as** hygiene products, groceries, and notably pet food (**Annex 11, Annex 6, Annex 13**)**. Their shopping behavior reflects** structure and stability**, showing low sensitivity to price and a clear preference for quality. They are likely to live in urban environments and demonstrate a** wellness-oriented lifestyle**, with care directed both toward themselves and their pets.**

4.3.3 Cluster 2 - Students

This cluster comprises the **youngest (Annex 16) and most price-conscious consumers.** Their shopping patterns are **strongly oriented toward discounts and promotions (Annex 15)**, with low overall expenditure and a limited range of product categories (**Annex 14**). Their behavior appears opportunistic and value-driven, suggesting they may be **students or individuals in the**

early stages of their professional lives. Although they demonstrate low brand loyalty, they are highly responsive to well-tailored promotional strategies. This cluster also represents the customers that complain the most (**Annex 4**).

4.3.4 Cluster 3 - Extended Households

This cluster is composed of **older clients (Annex 16) who live in larger family units**, likely with **teenagers and children at home (Annex 2, Annex 3)**. Their early shopping (8h30) (**Annex 20**) can indicate a structured routine, perhaps before school drop-offs. They exhibit **the highest diversity of products purchases (Annex 14)**, indicating complex household needs, and they have been clients for a long period of time. Despite their age, they still are in a **very active lifestyle**, taking care of others. Their spendings is **high in groceries, meat and hygiene, but they show very little sensitivity to promotions**, suggesting they prioritize reliability over price.

4.3.5 Cluster 4 - Tech Enthusiasts

This cluster consists of **recent customers** (**Annex 18)** whose purcahases are primarily focused on **technology products (Annex 19)**, with minimal engagement in traditional categories such as groceries or hygiene. They tend to **shop later in the day** and exhibit **low overall product diversity**. Their behavior suggests a **digitally oriented and goal-specific profile**, likely composed of younger individuals.

4.3.6 Cluster 5 - Mature Independents

This group consists of **long-standing**, **high-values customers**. They normally have **few or no children at home**, but given their age and tenure is likely that many are parents that their kids have already moved out.

They spend across multiple categories, especially in **hygiene**, **pet food** and **groceries**. Their afternoon shopping time indicates a flexible routine, possibly aligned with retirement or part-time work schedules.

Their **moderate promotion sensitivity** points to a balanced approach, they appreciate value but don't chase promotions, their buying is likely driven by convenience and comfort.

4.3.7 Cluster 6 - Makro Lovers

This **geographically** defined cluster consists of long-standing customers located **near the Makro Alfragide store**. They shop early in the day (around 9:15 a.m.), suggesting a habitual or time-efficient routine. These customers display **high cumulative spending**, particularly in groceries, alcohol, fish, and meat, yet show little engagement in categories such as hygiene, pet food, or technology.

Their basket diversity is moderate, with an average of only two distinct stores visited (**Annex 5**), and they demonstrate very low responsiveness to promotions. The absence of children or teenagers at home further supports a profile of mature, possibly independent individuals. Although they represent a spatially limited segment, their consistent patterns and high value make them well suited for location-based targeting and retention strategies.

4.4 Additional Comparisons

Loyalty Program Participation varies substantially by segment. Cluster 5 (Mature Independents) and Cluster 1 (Wellness Urbanites) show the highest share of loyalty card holders, suggesting strong brand attachment and habitual shopping patterns. Conversely, Cluster 4 (Tech Enthusiasts) and Cluster 2 (Students) demonstrate lower participation rates, reinforcing their opportunistic or purpose-driven behavior (**Annex 21**). These findings suggest that promotional strategies and loyalty incentives may be more effective when tailored to these differences in commitment.

Gender Distribution appears relatively balanced across clusters, with only slight variations. Males make up a slightly larger proportion in Cluster 3 (Extended Households) and Cluster 0 (Vegetarians), while females slightly dominate in Clusters 1 and 5. Although not a major differentiator, this consistency confirms that segmentation was driven more by behavioral and transactional variables than by gender. The Makro Lovers cluster demonstrates a balanced gender distribution, with no marked predominance of male or female customers. However, a notable deviation emerges in loyalty behavior - only approximately **10%** of this segment possess a loyalty card, placing them significantly below the average observed across other clusters (**Annex 22**).

Together, these comparative insights help clarify which clusters are more responsive to loyalty schemes and which may require alternative engagement strategies, such as targeted campaigns or personalized offers.

5 Basket Analysis and Customer Behavior

To enrich the segmentation results, transactional data from customer baskets is analyzed to uncover **purchasing patterns and associations**. This section explores the use of association rule mining to identify product affinities within each client segment, offering deeper behavioral insights that support targeted marketing efforts.

5.1 Association Rules

This section presents an association rule mining analysis conducted across distinct customer clusters to uncover patterns of co-purchased items. By identifying **frequent item sets and evaluating confidence and lift metrics**, the analysis reveals underlying relationships that are not immediately apparent from individual transactions. These insights are critical for informing data-driven strategies in product placement, bundling, and targeted promotions. A train-test split was implemented to ensure model robustness and to validate the generalizability of observed patterns. Association rules were extracted at varying confidence thresholds to capture both strong and emerging relationships across the different customer segments.

5.1.1 Vegetarian Customer Cluster

For the vegetarian cluster, frequent item sets (support ≥ 0.5) reveal **tomatoes** (76.87%), **asparagus** (64.36%), and **carrots** (54.65%) as core staples, with notable pairings such as asparagus and tomatoes (52.76%).

The presence of items like chicken may reflect data noise or broader customer definition. Association rules (confidence ≥ 0.2) show **strong relationships**, notably avocado → asparagus

(69.76%), carrots -> asparagus (69.79%), and cauliflower → asparagus (71.69%), while reciprocal rules confirm mutual associations, especially between asparagus and carrots.

High-lift rules (support ≥ 0.02) such as cauliflower and frozen vegetables → asparagus and carrots (lift: 1.42) suggest a **tendency to combine fresh and frozen produce**.

Test set validation (support ≥ 0.05) confirms the robustness of patterns, with rules like asparagus and green beans → tomatoes and mashed potato (lift: 1.32) recurring.

5.1.2 Wellness Urbanites Customer Cluster

The analysis of the "Wellness Urbanites" cluster reveals that **champagne is the most frequently purchased item**, appearing in over 75% of transactions, followed by fresh tuna at approximately 54%. Although just below the 0.5 support threshold, **Bluetooth headphones remain a significant item within this group**. Strong association rules indicate that customers who purchase almonds or asparagus are highly likely to also buy champagne and fresh tuna, with confidences reaching up to 81% for almonds leading to champagne. More complex item combinations, such as frozen smoothies paired with fresh tuna or spaghetti, suggest specific meal preparation patterns. High-lift rules highlight less obvious but meaningful relationships, including the tendency for buyers of cottage cheese and spaghetti to also purchase champagne and avocado.

Validation on the test set confirms **robust bidirectional associations** between technology products like laptops and Bluetooth headphones and luxury food items, reflecting a lifestyle that integrates wellness, and tech-savviness.

Overall, this cluster exhibits a sophisticated, health-conscious demographic that combines premium indulgences with nutritious foods and technological accessories.

5.1.3 Students Customer Cluster

For the student cluster, frequent items (support ≥ 0.5) reveal core staples such as **oil** (72.5%), **cooking oil** (51.6%), and **cake** (39.5%), alongside recurring items like **tea** (34.8%), **candy bars**, and **gum**. These patterns suggest a strong emphasis on **low-cost essentials and convenience-focused goods**, consistent with the profile of price-sensitive and time-constrained individuals.

Association rules (confidence ≥ 0.2) highlight notable co-purchase behaviors, including cake → candy bars (confidence: 35%) and asparagus → oil (confidence: 79%), the latter indicating occasional recipe-driven cooking behavior. Multi-item rules such as cooking oil + muffins → tea (confidence: 55%, lift: 1.51) and gum + cooking oil → tea (confidence: 56%, lift: 1.61) reflect an affinity for combining basic ingredients with comfort or study-related items. The high lift observed in alcohol-related rules - particularly white wine → cider (lift: 6.77) - suggests social or event-based purchasing patterns.

Test set validation confirms the persistence of these associations, with key rules (e.g., cooking oil and muffins → tea) maintaining their relevance, supporting the robustness of the patterns across subsets of data.

5.1.4 Extended Household Customer Cluster

For the extended household cluster, frequent item sets (support ≥ 0.5) indicate that baby's food (78.66%) and Ratchet & Clank (55.82%) are particularly common, with frequent co-occurrence of other child- and family-oriented products such as Minecraft (42.9%) and burgers. The presence of these items suggests a customer profile associated with **multi-member or family-based households**, potentially including both young children and adolescents.

Association rules (confidence ≥ 0.2) further highlight strong relationships such as asparagus → babies' food (confidence: 79.3%) and bacon → babies' food (confidence: 84.1%), indicating that these items may commonly appear in shared household baskets. Complex rules like Ratchet & Clank + Minecraft → Ratchet & Clank 2 + babies' food (lift: 1.20) reveal notable connections between gaming and parenting-related purchases, suggesting that households may cater to multiple age groups simultaneously. The coexistence of body spray, burgers, and gaming products with baby-related goods strengthens the assumption that these customers represent intergenerational or care-giving households.

High-lift rules (support ≥ 0.02) also support this interpretation, particularly combinations involving babies' food + Ratchet & Clank 2 + pet food, which consistently appear across the top-ranked associations. These patterns point to households managing multiple domestic needs—ranging from child care to pet ownership and entertainment—within the same shopping session.

Test set validation confirms the persistence of these associations, with rules such as **cooking oil + Minecraft → burgers and Ratchet & Clank + scallops → babies' food + Minecraft** maintaining high lift and confidence scores. These recurring patterns suggest a reliable and stable behavioral structure within this customer segment.

5.1.5 Tech enthusiast Customer Cluster

Regarding the tech enthusiast segment, the most frequently purchased items (support ≥ 0.05) include **energy drinks** (46.6%), **pancakes** (34%), and **white wine** (32%). These products seem to represent a blend of **quick energy, casual meals, and lifestyle-driven purchases**.

Looking at associations between products, an interesting rule emerges: 67.7% of the time that someone buys AirPods, they also purchase an energy drink (AirPods → Energy drink). This suggests a strong behavioral link between tech purchases and energy-boosting consumables.

Another compelling pattern supports this tech lifestyle: Shoppers who purchase both a gadget for TikTok streaming and a protein bar are 53.4% likely to also buy pancakes and energy drinks (confidence: 53.4%, lift: 1.98). This points to a clear "morning routine" or wellness-influencer shopper profile, where tech-savvy customers combine nutrition with gadgets, possibly for content creation, or productivity.

In the broader product space, high-lift rules (support ≥ 0.05) reveal further patterns. For instance, shoppers who buy both champagne and cider are 2.68 times more likely (lift: 2.68) to also buy white wine than a typical customer. This reflects a **celebratory buying pattern**, where consumers appear to be assembling drink assortments for social occasions.

5.1.6 Mature Independent

In this segment, the most frequently purchased items (support ≥ 0.5) include **oil** (69.2%), **barbecue sauce** (47.1%), and **cologne** (37.4%). These products indicate a household-focused basket, blending core cooking ingredients with personal care items.

Looking at the association rules (confidence ≥ 0.2), clear co-purchase relationships emerge. For example, when customers buy antioxidant juice, 59.4% of the time they also purchase barbecue sauce. Similarly, antioxidant juice strongly predicts the purchase of oil, with a confidence of 81.6%. These patterns reveal an interest in combining health-conscious beverages with practical cooking items potentially pointing to a segment that mixes wellness awareness with conventional meal prep habits.

Test set validation confirms that many of these rules persist across different subsets of the data. Rules such as barbecue sauce + cologne + oil → chicken and chicken + oil → barbecue sauce + cologne maintained high lift and confidence, reinforcing the **robustness of the family meal preparation and household care pattern**.

5.1.7 Makro Lovers

Viewing the Makro Lovers segment the most frequent items purchased (support ≥ 0.05) were **oil** (79%), **meatballs** (54,6%) and **mineral water** (48.4%). These items are essential, low-risk purchases consistent with Makro's reputation as a **bulk destination** for kitchen basics and shelf-stable goods. The pairing of oil and mineral water (38.4%) reinforces this, as both are kitchen essentials that are often bought together in high quantities.

Some examples of high lift association rules (support≥ 0.02), Cider + Oil → Eggs + Meatballs (lift: 6.36) this pairing indicates a meal prep planning, where ingredients are bought together.

Additional rules such as Mayonnaise + Mineral Water → Salmon (lift: 14.5) and Cake + Oil → Catfish, Vegetables Mix (lift: 9.66) further reflect comprehensive basket planning, not spontaneous purchasing. These shoppers are **planning, stocking and preparing in advance, which opens some opportunities for deals regarding meal ingredients**.

6 Targeted Promotions and Marketing Strategy

Based on the identified customer segments and basket analysis, this section proposes personalized promotional strategies aimed at maximizing engagement. Promotion mechanics such as **discounts, bundling, and loyalty incentives** are considered, along with an evaluation of their potential return on investment and business feasibility.

6.1 Segment-based Promotion Design

6.1.1 Vegetarian Customer Cluster

Bundled promotions can be offered for **high-frequency combinations** such as tomatoes, asparagus, and carrots, encouraging customers to purchase these staples together

In-store or online placement strategies should consider **positioning related items in proximity** -for example, placing avocados near asparagus, or organizing frozen vegetables alongside fresh produce like cauliflower. There is also an opportunity to prominently feature and promote vegan-related products to better cater to this customer segment.

Recipe-based marketing can highlight common pairings found in the data. **Distributing flyers with vegetarian and vegan recipes near the fresh produce section** can inspire meal planning and encourage customers to purchase complementary ingredients.

6.1.2 Wellness Urbanites Customer Cluster

Promotions that highlight frequently purchased combinations such as almonds, asparagus, champagne, and fresh tuna can encourage customer to buy these items together. Offering themed product collections, like a **"Healthy Pack" featuring champagne, Bluetooth headphones, and wellness items such as yoga mats**, can appeal to the lifestyle preferences of this customer segment and resonate with those who follow wellness trends such as yoga. Additionally, **cross-promoting** champagne and avocado with cottage cheese and spaghetti can leverage common meal pairing and drive complementary sales.

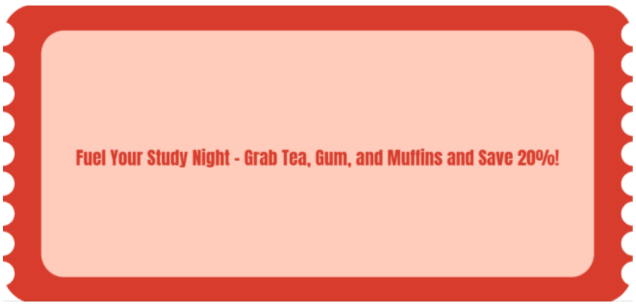
6.1.3 Students Customer Cluster

Promotional bundles can be built around frequently co-purchased items such as oil, cake, and tea, or cake and candy bars, reflecting the student segment's preference for affordable, convenient, and indulgent products. These combinations align with typical consumption habits and appeal to students' practical, comfort-driven shopping behavior.

There's also potential to offer simple **meal prep kits** using common pairings like cooking oil, frozen vegetables, and pasta, catering to students who often seek quick, low-effort meal solutions.

In addition, **strong associations between alcoholic beverages** like wine and cider suggest an opportunity for **event-based promotions**, particularly around weekends or campus events, provided they follow responsible marketing standards.

Lastly, **campaigns targeting late-night or study-time** consumption could spotlight items like tea, gum, and muffins. Framing these as part of a study essentials collection may boost their appeal and reinforce the retailer's relevance in student life.



6.1.4 Extended Household Customer Cluster

Marketing strategies for the extended household segment should emphasize **cross-category bundles that reflect their diverse domestic needs**. Promotions combining items like baby food, cooking oil, and burgers, or video games with snacks and child-care products, align well with this group's buying behavior.

Product placement can be optimized by positioning children's food near parent-focused or family entertainment items, for example, pairing baby snacks near gaming displays or frozen meals. This encourages basket expansion and supports a smoother shopping experience.

Given the presence of **multi-generational or caregiving households, curated "Family Night Bundles"** featuring products like Ratchet & Clank, burgers, and desserts, or quick-prep dinner kits tailored for children and teens, can drive larger purchases while boosting brand relevance.

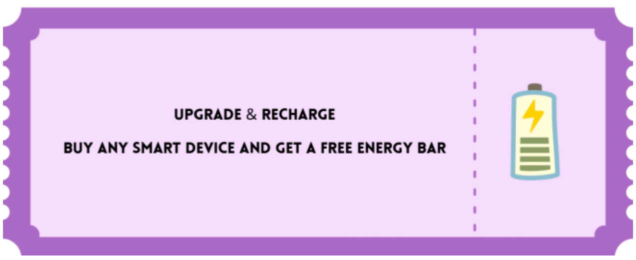
Additionally, the consistent appearance of pet food in high-lift rules suggests an opportunity for pet-inclusive household promotions, positioning the retailer as a one-stop solution for diverse family needs.

6.1.5 Tech Enthusiast Customer Cluster

Promotions can be strategically designed by leveraging the frequent item combinations and behavioral patterns revealed in the data. High frequency groupings such as energy drinks, pancakes and protein bars are common buying habits, creating a **bundled promotion** based on these items can encourage tech enthusiasts to purchase these items together.

Products like **AirPods and energy drinks can be placed in proximity near checkout areas** or in **sections tailored to students or young professionals**. Similarly, gadgets associated with TikTok streaming can be displayed alongside protein bars or other quick-energy snacks in wellness or lifestyle aisles.

Beyond simple bundling and placement, these purchasing patterns also open the door for lifestyle-driven marketing campaigns.



6.1.6 Mature Independents

Custom promotions can be crafted around regularly purchased combinations such as oil, cologne, and barbecue sauce, which reflect the Mature Independent segment's focus on practicality, efficiency, and maintaining a well-stocked home. There's also a clear opportunity to introduce **ready-to-cook bundles** based on sets featuring barbecue sauce, chicken, mashed potatoes, ham and cooking oil, could serve individuals that are looking to prepare simple meals with minimal planning.

Additionally, frequent links between personal care products and groceries such as deodorant and chicken, or cologne and oil, suggest potential for **cross-category promotions**.

Lastly, **smaller comfort groupings**, such as cake, tea and personal care items could form an ideal set for shoppers who value quality time at home. These bundles would position the retailer as a source not only for basic needs but also for self-care moments.

6.1.7 Makro Lovers Customer Cluster

Promotional bundles for the Makro Lovers segment can be developed around **core essentials** such as oil, mineral water, and meatballs which are frequently co-purchased and reflect the typical bulk-buying behavior of Makro's customer base. These combinations suggest **practical, high-volume purchases likely tied to family meal planning, small business needs, or regular restocking**. Bundles like **"Kitchen Basics in Bulk"** could cater to this segment by offering savings on key pantry staples.

There's also potential to create recipe-based kits designed for meal preparation, based on high-lift associations.

Finally, beverage-driven rules, such as white wine, cider, and mineral water support the creation of event-based or restock-focused promotions, especially relevant for hospitality clients or large households. These promotions could be activated seasonally or around holidays, emphasizing convenience, cost-efficiency, and Makro's advantage in bulk availability.

6.2 Campaign Feasibility Analysis

To ensure each proposed promotion achieves maximum impact while maintaining operational and financial viability, we evaluate the feasibility of campaign strategies across three core dimensions:

- **Customer Engagement Potential**: Likelihood of the promotion to resonate with the target segment.
- **Operational Feasibility**: Practicality of implementation, including inventory, supply chain, and merchandising.
- **Return on Investment (ROI)**: Expected revenue uplift relative to promotional costs.

Cluster	Engagement Potential	Operational Feasibility	ROI Expectation
Vegetarian	**High.** Health and eco-conscious consumers are receptive to recipe-based promotions and bundling of plant-based items.	**High.** Fresh produce and vegan alternatives are staple inventory items.	**Moderate to High.** While margins on fresh produce are lower, bundling encourages higher basket sizes.
Wellness Urbanites	**Very High.** Lifestyle-oriented bundles and premium wellness promotions align closely with values.	**Moderate.** Requires coordination across departments (e.g., electronics + food + wellness).	**High.** Premium pricing and lifestyle appeal can drive strong conversion and brand loyalty.
Students	High. Students respond well to value-driven and convenience-focused promotions.	High. Most products (tea, gum, pasta) are existing stock and easy to bundle.	Moderate to High. Low cost-to-serve and high-volume potential, especially around university semesters.
Extended Household	**High.** Large households appreciate value and convenience from cross-category bundles.	**Moderate.** Requires broad coordination across product teams (children's goods, snacks).	**High.** Large basket sizes and recurring needs make this segment particularly valuable.
Tech Enthusiast	**Very High.** Segment responds to lifestyle alignment and digital engagement.	**Moderate.** Needs coordination between electronics and grocery categories.	**High.** Premium product attachment and high engagement levels justify marketing spend.
Mature Independents	**High.** Practical bundles (oil, cologne) and comfort sets (cake, personal care) align with this group's home-focused lifestyle.	**High.** Cross-category bundling (groceries + personal care) is manageable.	**Moderate to High.** Ready-to-cook and self-care kits can drive both basket size and brand affinity.
Makro Lovers	**Very High.** Bulk bundles match this	**Very High.** Makro's setup supports volume	**High.** Large basket sizes and frequency

	group's restocking and value-driven habits.	deals with minimal changes.	make these promos profitable.

7 Final Git Repository Organization

This section outlines the structure of the project's *Git* repository, detailing the categorization of files into datasets, *Jupyter* notebooks, and *Python* scripts to ensure clarity, reproducibility, and maintainability of the analytical workflow.

7.1 Datasets

- ***customer_info.csv*** - Raw dataset containing customer demographic and behavioral attributes used as the base for preprocessing.
- ***customer_basket.csv*** - Raw transactional dataset listing product purchases grouped by customer ID, used for market basket analysis.
- ***makro_cluster.csv*** - Dataset containing customer records manually identified as part of the Makro Alfragide geographic cluster.
- ***outlier_dataset.csv*** - Dataset of outlier records identified and removed prior to clustering, later re-integrated via centroid assignment.
- ***info_clustering.csv*** - Cleaned and feature-engineered dataset used for clustering, after preprocessing, feature selection, and outlier removal.
- ***dataset_cluster_final.csv*** - Final dataset combining cluster labels from K-Means and outlier assignments via nearest-centroid method.
- ***id_and_cluster.csv*** - Simplified mapping file linking each customer ID to its assigned cluster, used in downstream analyses such as basket segmentation.

7.2 Notebooks

- ***notebook.ipynb*** - Main preprocessing pipeline including missing value handling, feature engineering, and data transformations.
- ***Clustering.ipynb*** - Implementation and evaluation of clustering algorithms (K-Means, Hierarchical, SOM), including visual diagnostics.
- ***som.ipynb*** - Analysis notebook for Self-Organizing Maps (SOM), used to extract insights and visualize quantization error distributions.
- ***Cluster_characterization.ipynb*** - Exploratory notebook used to generate bar plots, value counts, and descriptive analyses per cluster.
- ***basket_analysis.ipynb*** - Executes segment-wise association rule mining using Apriori, including rule interpretation and visualization.

7.3 Python Scripts

- **info_preprocessing.py** - *Functional version of the preprocessing pipeline, with reusable functions for cleaning, transforming, and exporting data.*
- **Clustering.py** - *Modular implementation of clustering logic, including grid search, evaluation metrics, and visualization helpers.*
- **som.py** - *Script containing functions to run and analyze SOMs independently of the notebook environment.*

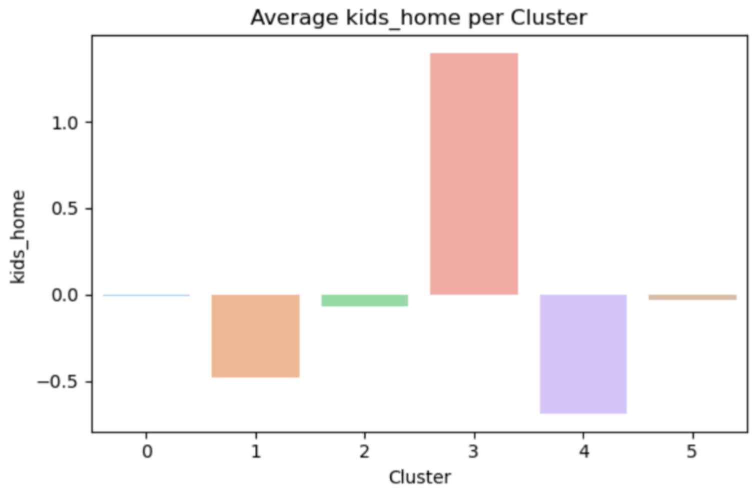
- **Clustering_characterization.py** - *Functions to support cluster interpretation via value counts, feature summaries, and visual comparisons.*
- **Basket_analysis.py** - *Scripted version of the basket analysis process, including functions for frequent itemset mining and rule evaluation.*

8 Conclusion

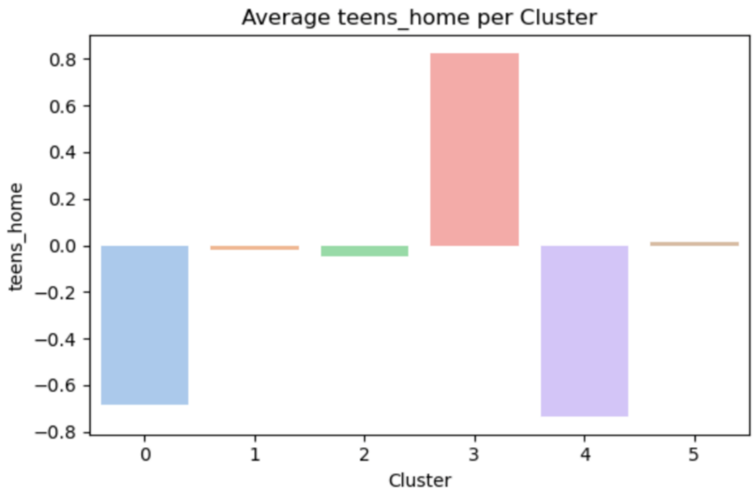
This study demonstrates the effectiveness of unsupervised learning for uncovering meaningful customer segments based on diverse behavioral, demographic, and transactional features. The combination of clustering methods and basket analysis revealed interpretable patterns in consumer behavior, enabling the design of targeted marketing strategies aligned with each segment’s profile. The integration of spatial analysis further enhanced segmentation by capturing geographic nuances. Overall, the results underscore the strategic value of data-driven segmentation for personalizing promotions, optimizing customer engagement, and informing retail decision-making.

9 Annexes

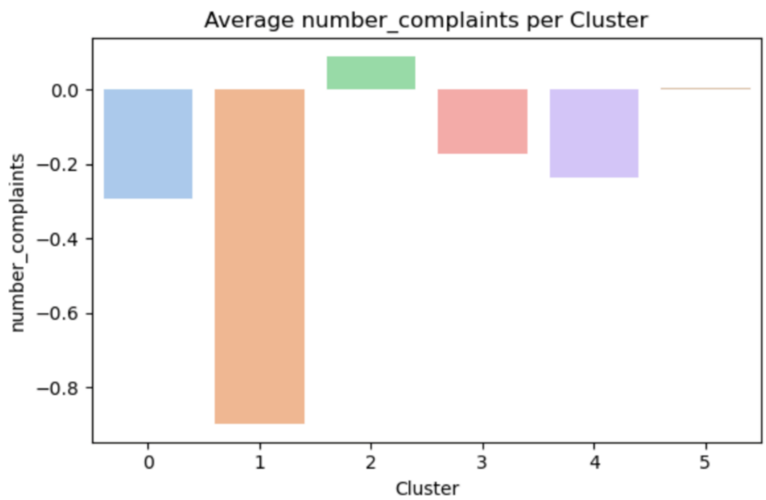
Annex 1 – Histograms and distributions of 'info_dataset_copy'



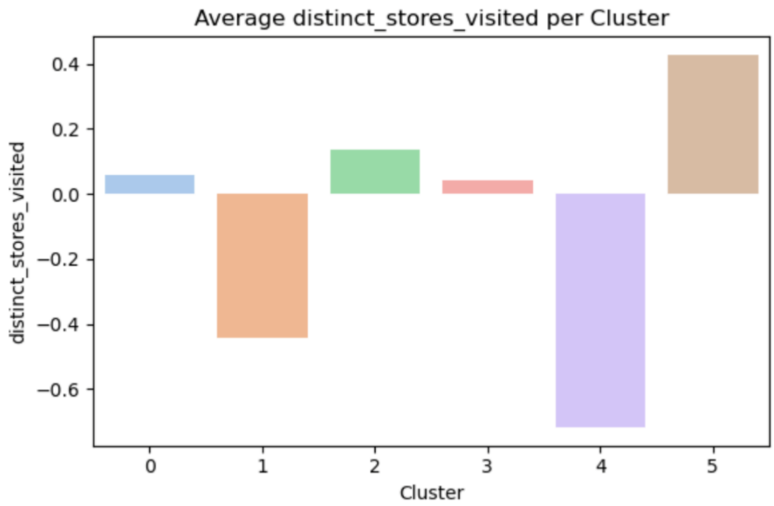
Annex 2 – Bar Plot of Average kids_home per Cluster



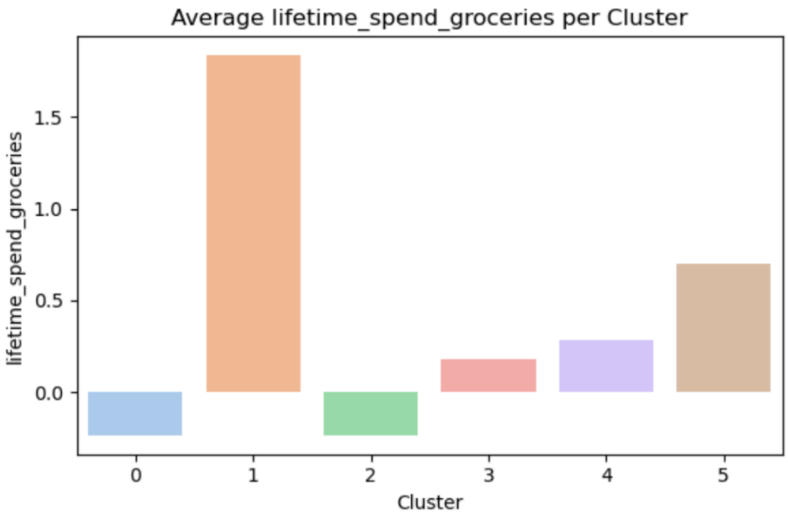
Annex 3 - Bar Plot of Average teens_home per Cluster



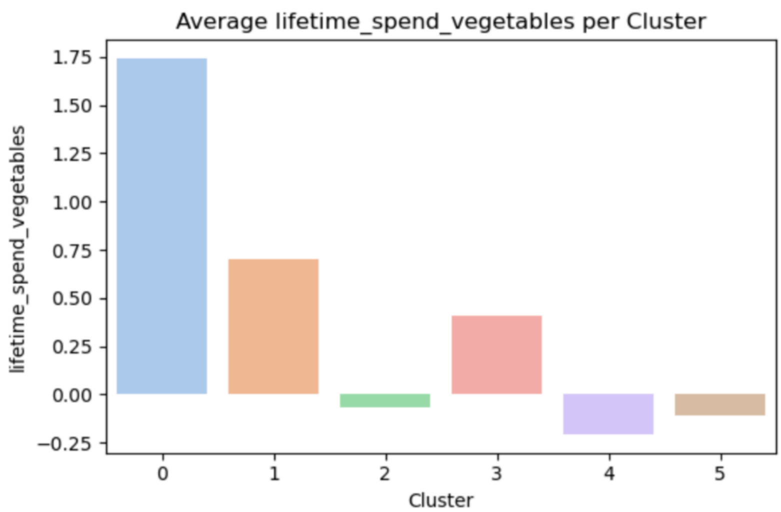
Annex 4 - Bar Plot of Average number_complaints per Cluster



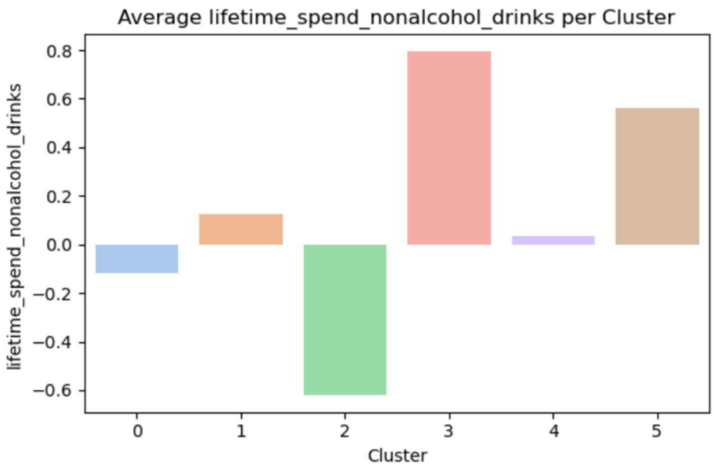
Annex 5 - Bar Plot of Average distinct_stores_visited per Cluster



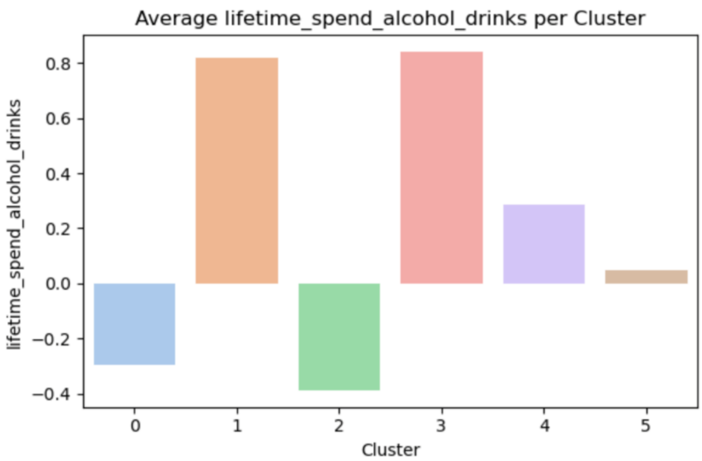
Annex 6 - Bar Plot of Average lifetime_spend_groceries per Cluster



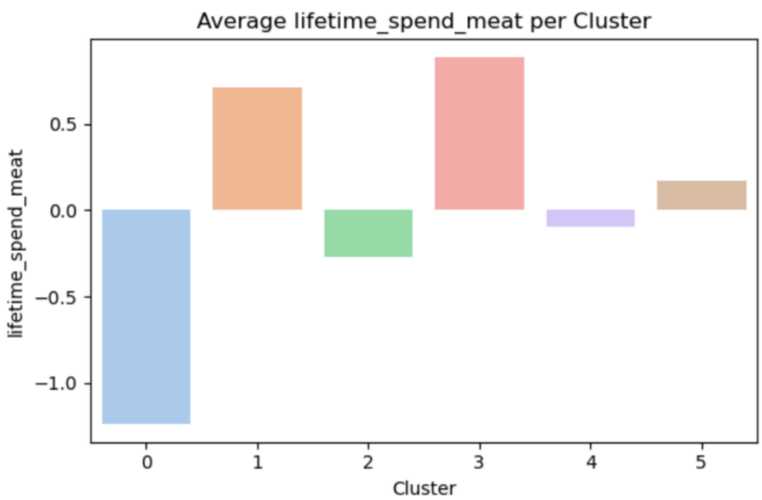
Annex 7- Bar Plot of Average lifetime_spend_vegetables per Cluster



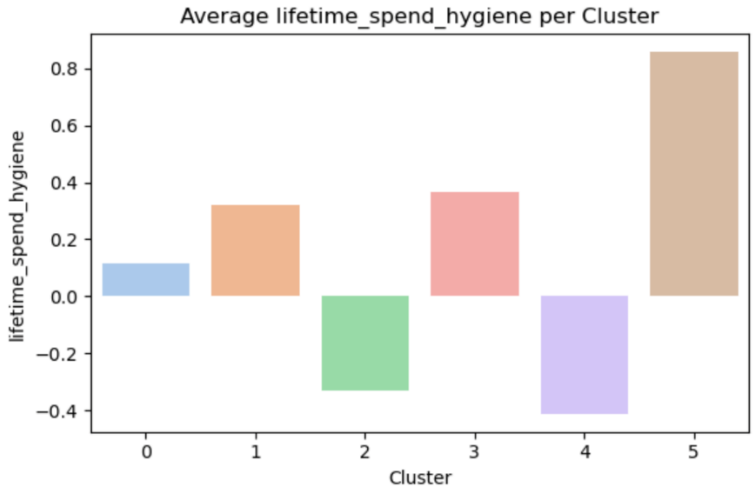
Annex 8- Bar Plot of Average lifetime_spend_nonalcohol_drinks per Cluster



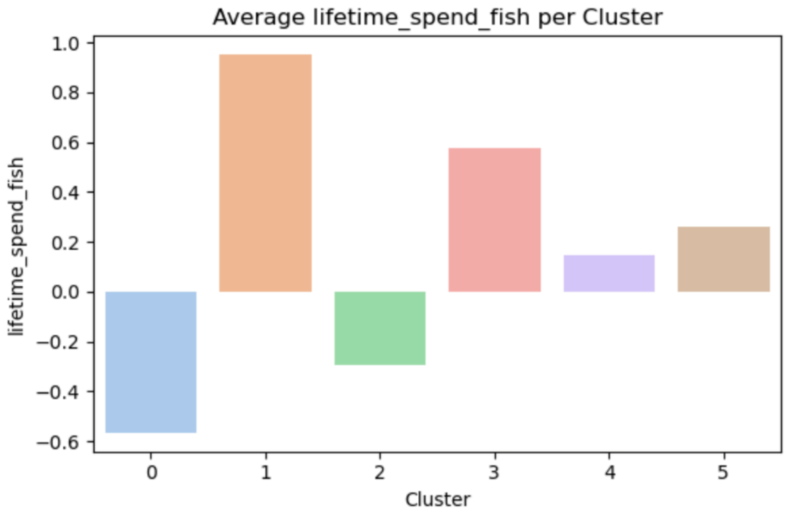
Annex 9- Bar Plot of Average lifetime_spend_alcohol_drinks per Cluster



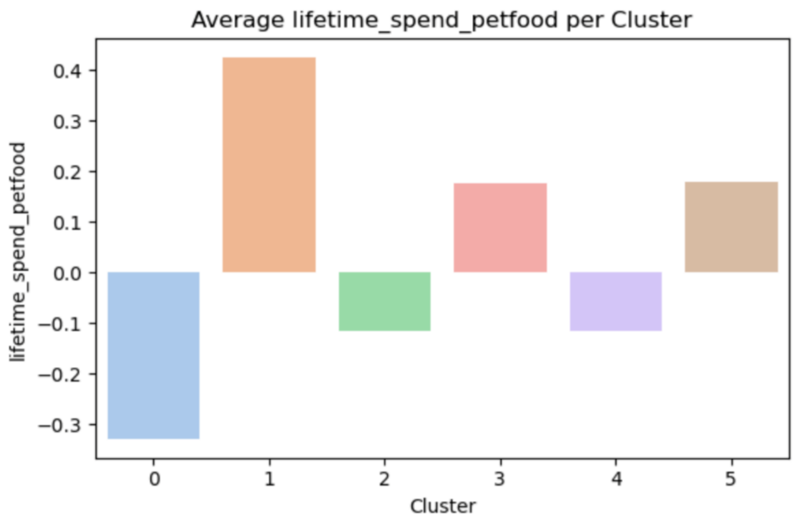
Annex 10- Bar Plot of Average lifetime_spend_meat per Cluster



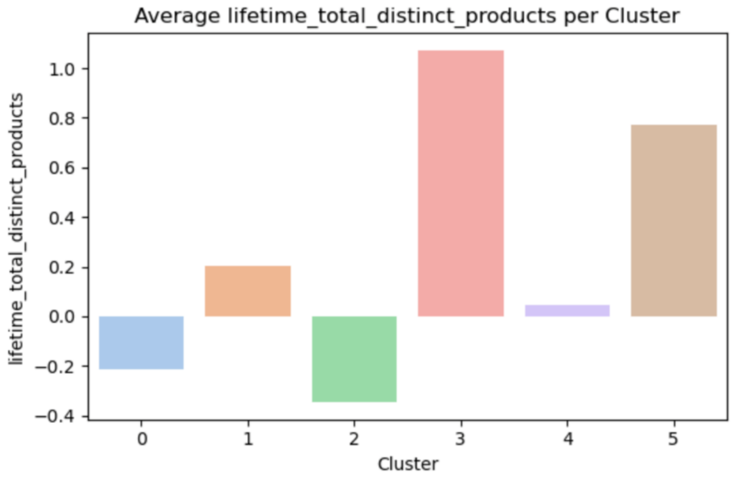
Annex 11- Bar Plot of Average lifetime_spend_hygiene per Cluster



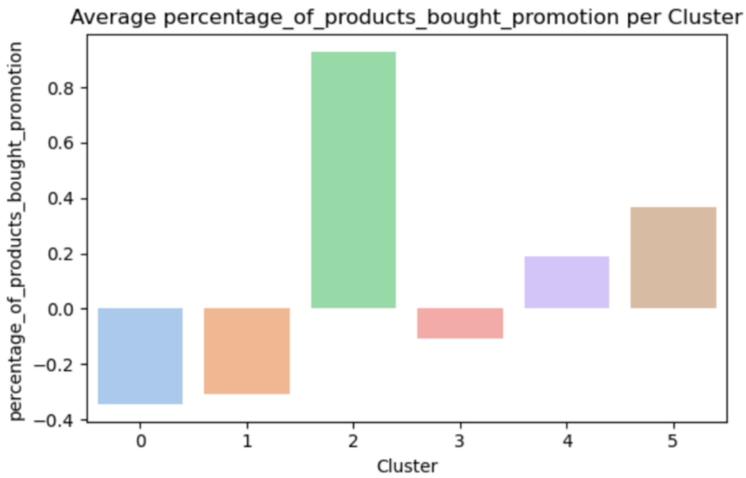
Annex 12- Bar Plot of Average lifetime_spend_fish per Cluster



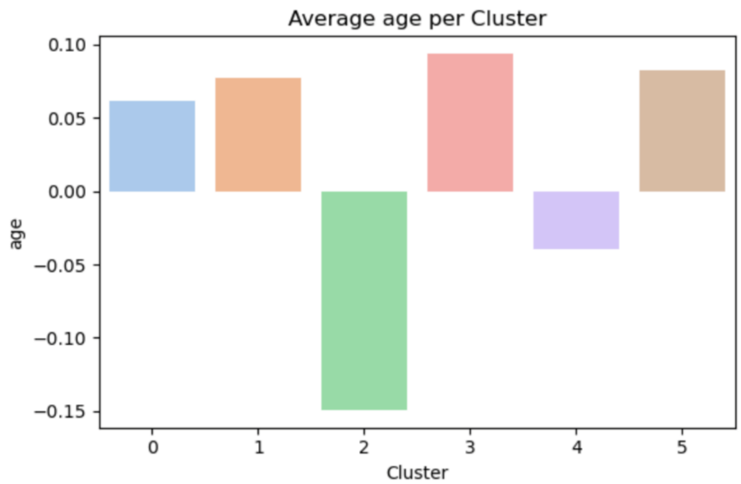
Annex 13- Bar Plot of Average lifetime_spend_petfood per Cluster



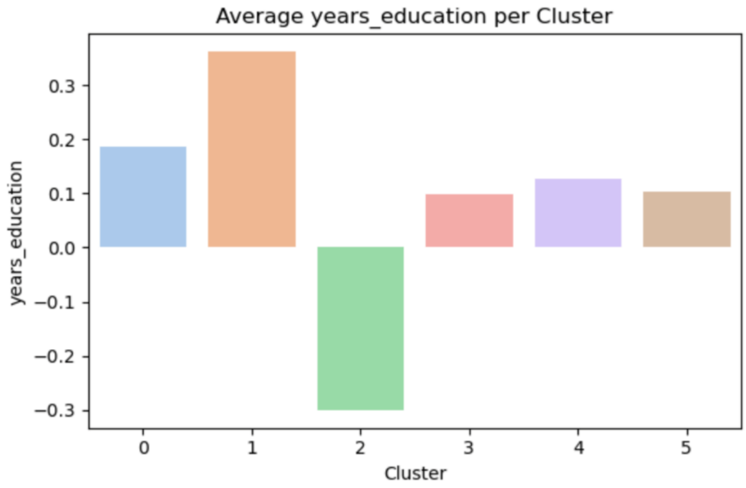
Annex 14- Bar Plot of Average lifetime_total_distinct_products per Cluster



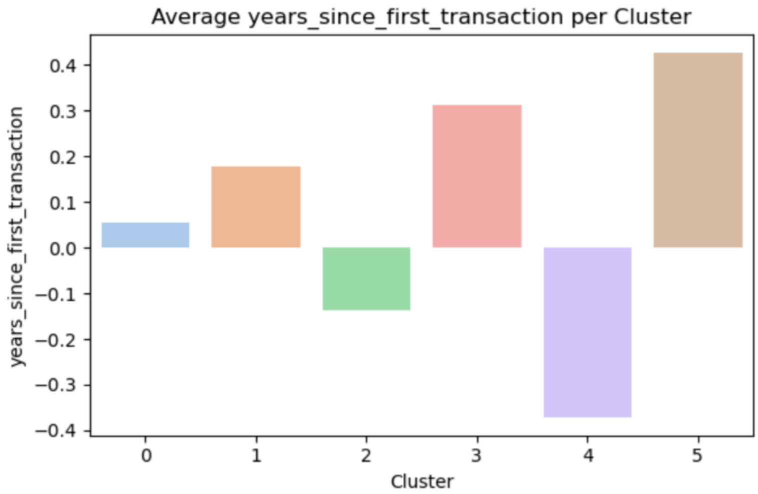
Annex 15- Bar Plot of Average percentage_of_products_bought_promotion per Cluster



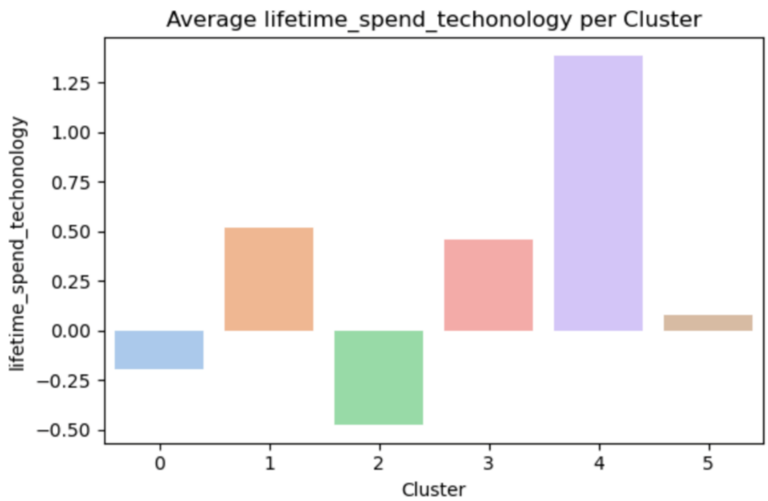
Annex 16- Bar Plot of Average age per Cluster



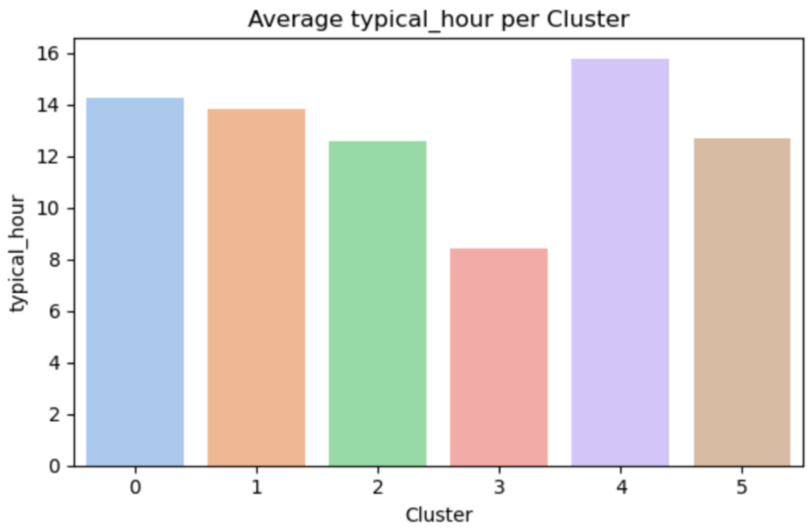
Annex 17- Bar Plot of Average years_education per Cluster



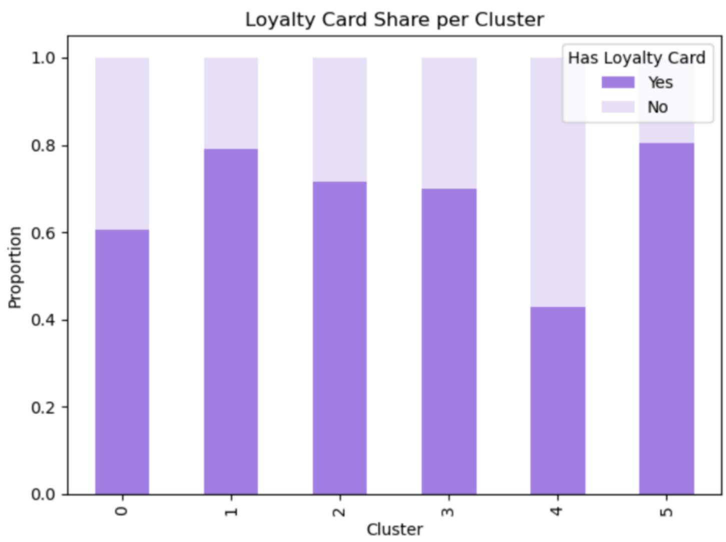
Annex 18- Bar Plot of Average years_since_first_transaction per Cluster



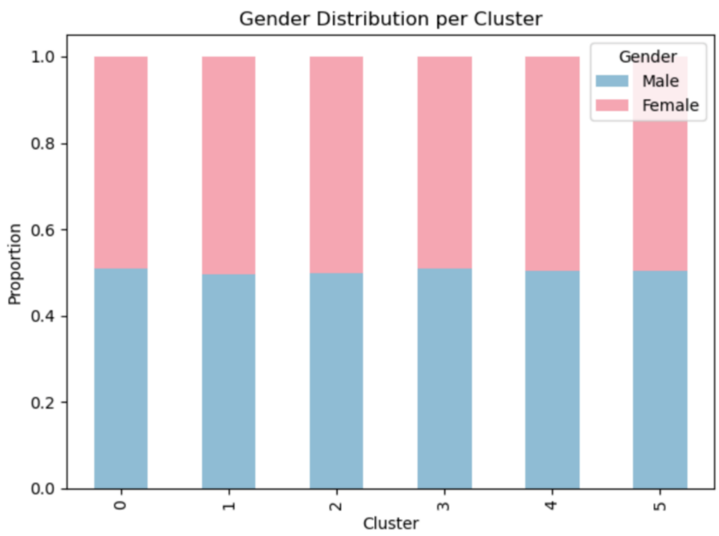
Annex 19- Bar Plot of Average lifetime_spend_technology per Cluster



Annex 20- Bar Plot of Average typical_hour per Cluster



Annex 21- Stacked Bar Chart about Loyalty Card Share per Cluster



Annex 22- Stacked Bar Chart about the Gender Distribution per Cluster